Blockchain Application Engineer — *Smart Contract*s & *Web3* Integration

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Department: *Blockchain* & Software Engineering
Reports To: Lead *Blockchain* Engineer / Head of Engineering
Level: Mid-Level / Senior

About the Company

We are preparing to establish an independent software development company in Poland, focusing on building secure, scalable financial technology and *blockchain* applications.

Our planned development areas include:

- **Digital Asset Platforms:** *Cryptocurrency* transactions, *wallet*s, asset management, and *blockchain* integrations.
- **Decentralized Finance (*DeFi*):** *Trading* applications, staking, lending, liquidity management, and decentralized exchanges.
- **Fintech & Payments:** Stable*coin* transfers, payment processing, financial APIs, and settlement infrastructure.
- ***Web3* Applications:** *Wallet*-connected platforms, *smart contract* integrations, and decentralized application development.
- **Real-World Asset *Token*ization:** *Blockchain*-based ownership records, *token*ized assets, and financial verification.

We aim to build reliable applications that connect traditional software systems with decentralized technologies.

Our engineering team will work across frontend, backend, *smart contract*s, and cloud infrastructure, with a s*trong* emphasis on security, maintainability, and production reliability.

About the Role

We are looking for a ***Blockchain* Application Engineer — *Smart Contract*s & *Web3* Integration** to design, build, and maintain *blockchain*-connected applications.

The role combines hands-on *smart contract* development with frontend and backend integration.

You will work with *Solidity*, *Ethereum*-compatible networks, *Web3* libraries, APIs, and modern application frameworks to create secure and reliable financial products.

You will collaborate with frontend, backend, *blockchain* infrastructure, and security engineers to support complete transaction workflows, from user interactions and *wallet* connections to *smart contract* execution and transaction confirmation.

This position is well suited to an engineer who enjoys practical *Web3* development and wants to contribute across multiple layers of *blockchain* applications.

What You'll Do

Smart Contract Development

- Design, implement, test, and maintain *Solidity smart contract*s.
- Develop contracts supporting *token*s, asset transfers, staking, rewards, and other financial workflows.
- Work with ERC-20, ERC-721, ERC-1155, and relevant *Ethereum* standards.
- Implement appropriate access controls, authorization mechanisms, and contract validations.
- Integrate established *smart contract* libraries such as OpenZeppelin.
- Write automated unit, integration, fuzz, and invariant tests where appropriate.
- Optimize *Solidity* code for correctness, maintainability, and reasonable gas costs.
- Support *smart contract* deployment, verification, configuration, and upgrades.

Web3 Application Integration

- Integrate *smart contract*s with frontend and backend applications.
- Build *wallet* connection and authentication workflows using *MetaMask*, *Wallet*Connect, and related technologies.
- Work with *eth*ers.js, viem, wagmi, or comparable *Web3* libraries.
- Implement transaction signing, submission, confirmation tracking, and error handling.
- Support *blockchain* network switching, *token* approvals, balances, and transaction history.
- Build interfaces for displaying *blockchain* transaction data and contract states.
- Integrate decentralized applications with *Ethereum* and *EVM*-compatible networks.
- Help create intuitive and reliable user experiences for *blockchain* transactions.

Backend & *Blockchain* Services

- Develop backend services connecting financial applications with *blockchain* networks.
- Build REST or GraphQL APIs for transaction processing, contract interactions, and financial data.
- Implement *blockchain* event listeners, transaction monitors, and in*dex*ing services.
- Handle transaction retries, failed submissions, confirmations, and chain reorganizations.
- Work with PostgreSQL, Redis, or comparable database and caching technologies.
- Develop asynchronous processing and background jobs for *blockchain*-related workflows.
- Integrate external RPC providers, *blockchain* explorers, and third-party financial services.
- Maintain consistency between off-chain application data and on-chain transaction states.

DeFi & Financial Application Development

- Integrate decentralized exchanges, *token* contracts, and relevant *DeFi* protocols.
- Support *token swap*s, staking, withdrawals, deposits, and reward distribution.
- Build *smart contract* interactions for digital asset management platforms.
- Implement safeguards for *token* approvals, slippage, transaction validation, and user permissions.
- Collaborate with backend engineers on financial transaction workflows.
- Support stable*coin* payment and *blockchain* settlement integrations.
- Participate in the development of secure *wallet* and portfolio management features.
- Understand and manage risks related to external protocols and third-party contracts.

Security & Testing

- Follow secure *smart contract* and application development practices.
- Identify and mitigate common vulnerabilities such as reentrancy, improper authorization, and unsafe external calls.
- Review contract assumptions, economic edge cases, and transaction lifecycle risks.
- Write automated tests using Foundry, Hardhat, or comparable tools.
- Participate in code reviews, security assessments, and remediation work.
- Apply secure key-management practices and avoid exposing sensitive credentials.
- Collaborate with security engineers and external auditors when required.
- Maintain clear documentation of important contract and integration behaviors.

Collaboration & Delivery

- Work with product managers and engineers to translate requirements into technical solutions.
- Participate in architecture discussions, implementation planning, and code reviews.
- Maintain readable, tested, and well-documented code.

- Investigate production issues involving *wallet*s, RPC services, APIs, and *smart contract*s.
- Contribute to CI/CD pipelines and deployment automation.
- Share knowledge and support less-experienced engineers.
- Take responsibility for features from implementation through release and maintenance.

Required Skills & Experience

- 3+ years of professional software development experience.
- Practical experience developing *blockchain* applications or integrating *smart contract*s.
- Proficiency in *Solidity* and a working understanding of the *Ethereum* Virtual Machine.
- Experience with *Ethereum* or other *EVM*-compatible *blockchain* networks.
- Familiarity with *smart contract* development tools such as Foundry or Hardhat.
- Experience integrating *smart contract*s using *eth*ers.js, viem, *Web3*.js, or similar libraries.
- S*tron*g JavaScript or TypeScript programming skills.
- Experience with React, Next.js, Node.js, or comparable web development frameworks.
- Understanding of *wallet* interactions, transaction signing, gas fees, and *blockchain* confirmations.
- Knowledge of REST APIs, asynchronous operations, and database fundamentals.
- Understanding of basic *smart contract* security principles.
- Experience with Git, testing, debugging, and code reviews.
- Ability to work independently and collaborate effectively in a remote engineering team.
- Good written communication and technical problem-solving skills.

Nice to Have (Differentiators)

- Experience building production *DeFi* applications.
- Knowledge of *Uniswap*, *Aave*, Curve, or comparable protocols.
- Experience with ERC-4626 vaults, staking, lending, or governance contracts.
- Practical experience with Foundry fuzz testing and invariant testing.
- Experience with *smart contract* audits or vulnerability research.
- Familiarity with *Solidity* proxy and upgradeability patterns.
- Experience with *wallet* infrastructure, account abstraction, or EIP-712 signatures.
- Knowledge of *blockchain* in*dex*ing tools, RPC/WebSocket APIs, or The Graph.
- Experience with Python, Go, *Rust*, C++, or C# backend services.
- Familiarity with stable*coin* payments and *token*ized financial assets.
- Knowledge of cross-chain messaging and bridge integration.
- Experience with AWS, Docker, Kubernetes, or CI/CD automation.
- Contributions to open-source *blockchain* projects.
- Experience working with distributed international engineering teams.

Education

- Bachelor's degree in Computer Science, Software Engineering, Information Technology, or a related field.
- Equivalent professional experience and practical *blockchain* development skills are also welcome.
- Relevant open-source contributions, deployed applications, and verifiable *smart contract* projects are highly valued.

What Success Looks Like

First 3 Months

- Understand the project's application architecture and *blockchain* integration requirements.
- Become familiar with existing development tools, repositories, and security practices.
- Deliver initial *smart contract* or *Web3* integration features.
- Contribute reliable automated tests and technical documentation.
- Collaborate effectively with frontend and backend engineers.

Within 6 Months

- Independently develop and maintain *blockchain*-connected application features.
- Deliver secure *smart contract* integrations across supported networks.
- Improve transaction handling, *wallet* workflows, and application reliability.
- Contribute to security reviews and production troubleshooting.
- Help standardize reusable *Web3* components and development practices.

Long-Term

- Take ownership of important *blockchain* application components.
- Improve *smart contract* security, testing, and maintainability.
- Support additional *blockchain* networks and financial product integrations.
- Contribute to scalable and reliable *Web3* application architecture.
- Help strengthen engineering standards as the development team grows.

Why Join Us?

- Help build financial and *Web3* applications at an early-stage independent software company.
- Work with *smart contract*s, *blockchain* infrastructure, and modern full-stack technologies.
- Contribute to digital asset platforms, decentralized exchanges, and payment applications.

- Take meaningful ownership of engineering features and technical solutions.
- Collaborate with an international, remote-friendly engineering team.
- Explore flexible full-time, part-time, or contract arrangements.
- Develop practical experience across both on-chain and off-chain application engineering.

Ideal Candidate

We are looking for a practical software engineer who enjoys developing *smart contract*s and connecting *blockchain* technology with real applications.

You should be comfortable writing *Solidity*, integrating contracts with web applications, working with *blockchain* APIs, and investigating transaction-related issues.

You do not need to be an expert in every blockchain ecosystem. We value strong software engineering fundamentals, security awareness, curiosity, and the ability to deliver reliable solutions.

The ideal Blockchain Application Engineer combines Solidity development, Web3 integration skills, backend fundamentals, and a strong understanding of secure blockchain transaction workflows.